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Intelligent CyberInfrastructure with Computational Learning in the Environment (ICICLE)



<https://icicle.ai>



ICICLE

Three type of Artificial Intelligence

- Intelligence is the ability to understand, learn, and/or adapt to new problems [adapted from Merriam Webster]
- Artificial intelligence (AI) is computer software that exhibits some aspect of intelligence
 - For example, AI may adapt to only certain problems
 - AI may not understand how it solves or problem or how that problem relates to bigger problems

Three types of Artificial Intelligence

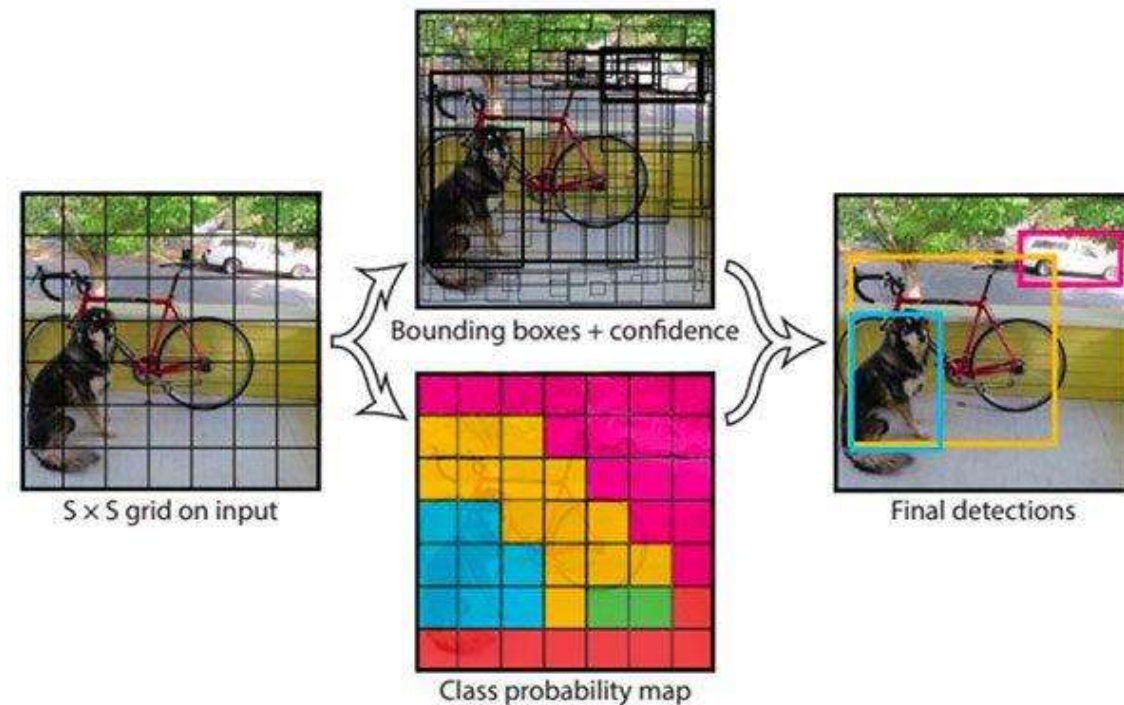
- ✓ Object detection
- ✓ Predictive
- ✓ Generative



Object Detection using YOLO

- YOLO = “You Only Look Once”
- It looks at the whole image just once to find all objects.
- YOLO’s trick:
 - Converts image into a grid.
 - Predicts what’s inside each grid cell in one go!

Result: Super-fast object detection,
great for drones, robots, and self-driving cars



Predictive AI

- Learns patterns from labeled data
- Uses math models (regression, decision trees, neural nets)
- Outputs a number, category, or probability
- Does **NOT** create new content

Examples of Predictive AI models:

- YOLO → Detects objects (classification + bounding boxes)
- ResNet / EfficientNet → Image classification
- XGBoost / Random Forest → Fraud detection, credit scoring

Key traits:

- Fast
- Lightweight
- Runs on edge devices
- Usually task-specific
- Needs structured, labeled data



Generative AI

- Learns from massive amounts of unstructured data
- Models use transformers, attention, and billions of parameters
- Generates **NEW** data, not just predictions

Examples of Generative AI models:

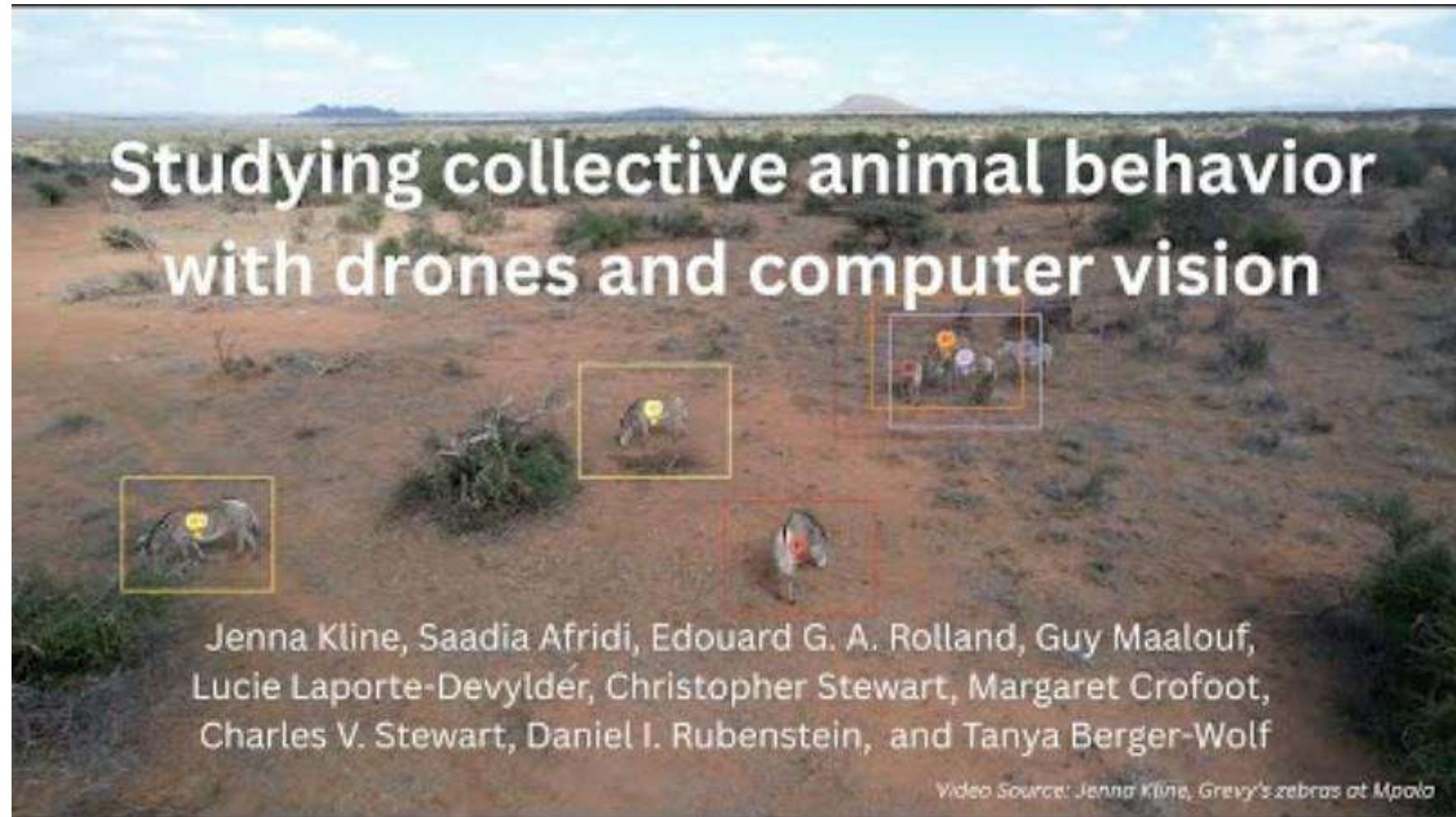
- ChatGPT(GPT-4 / GPT-5), Claude, Gemini, etc → chat, coding, writing
- LLaMA, Mistral → open-source LLMs

Key traits:

- Either too slow or may not work without GPUs
- Needs internet/cloud
- Memory-heavy
- Can hallucinate



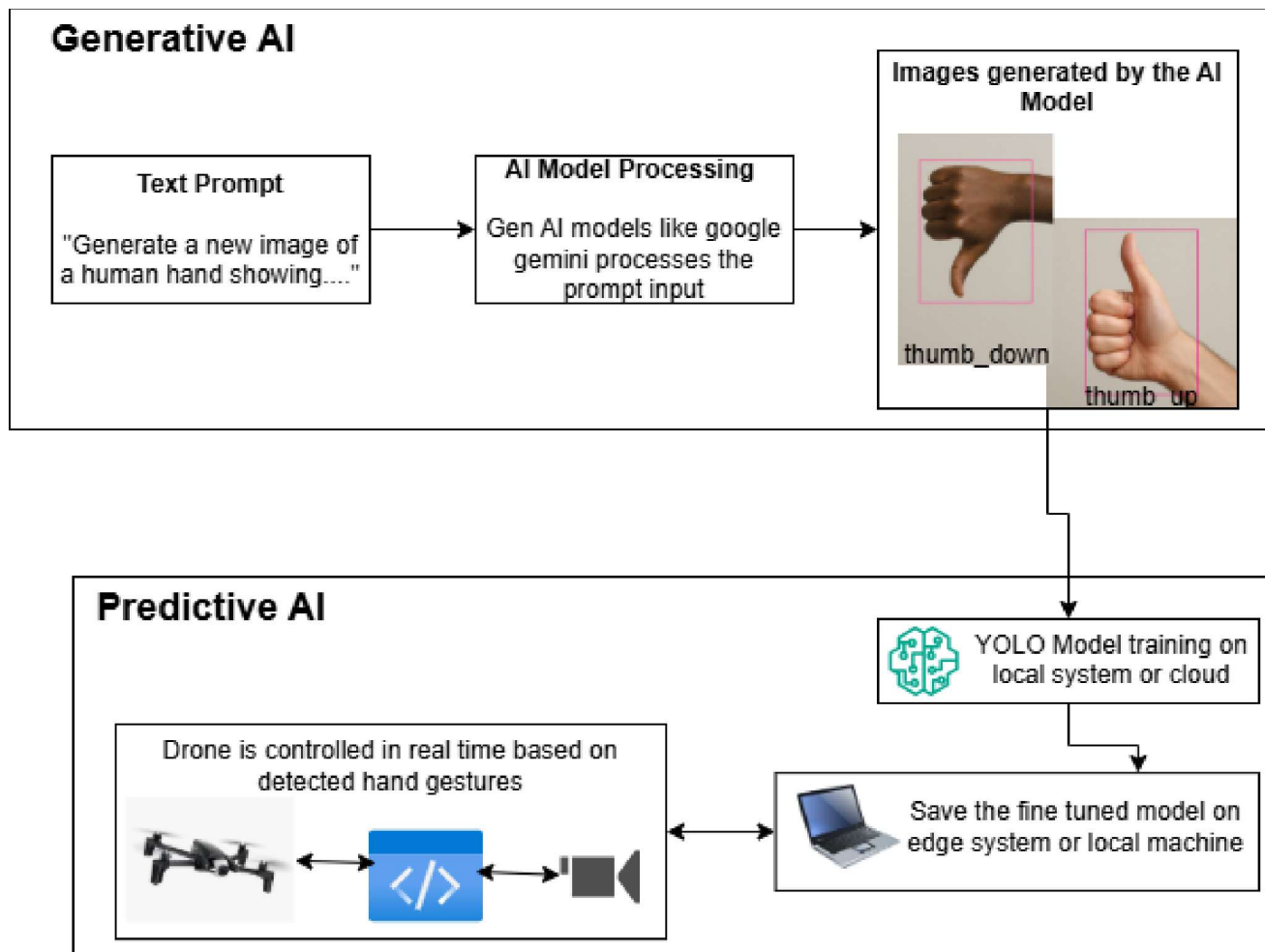
Using each type of AI to make the world better

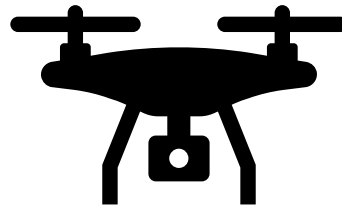


Collected with the Imageomics Institute, video editing by Ryan Horn and Jenna Kline (2025)



Hands-On Experience: Using Generative AI to train Predictive AI





Can we control drone using our hand gestures?
Let's Find Out!

